Subject Code: BMBIT03

PAPER ID-311807

Roll No: | | | | | | | | | | | | | |

MBA
(SEM III) THEORY EXAMINATION 2025-26
DATABASE MANAGEMENT SYSTEM

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question
a.	Define Data, Database & Database Management System.
b.	What do you mean by data model?
c.	Define Primary key with example.
d.	What is the syntax of Insert clause used in SQL?
e.	Describe relational calculus.
f.	What do you mean by Transaction in databases?
g.	Discuss data mining.

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Q no.	Question
a.	Describe evolution of database system.
b.	Write down the basics of relational model. How E/R diagrams can be converted to relation designs?
c.	Discuss the extended operators of relational algebra
d.	Explain the process of normalization up to BCNF with examples.
e.	Explain the "Two- Phase" locking protocol and its variants.

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question
a.	Differentiate between file-oriented system & database system.
b.	What is ER diagram? Draw a suitable ER diagram for a hospital management system.

4. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question
a.	Explain Generalization, Specialization and Aggregation with examples.
b.	What do you mean by ACID properties of a transaction? Discuss each of them.

5. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question
a.	What are the various Integrity Constraints? Discuss each of them.
b.	Describe various types of join operations. Briefly explain their applications.

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6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	What is Deadlock? Explain deadlock detection and prevention techniques
b.	Explain indexing. Differentiate between primary, secondary and clustered indexes.

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question
a.	Discuss data warehousing, OLAP & OLTP.
b.	Explain database security briefly.

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